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aiphoria: A Python package for dynamic wood material flow and carbon stock analysis

Timokleia Orfanidou [1] **and Janne Järvikylä** [1]

1 European Forest Institute (EFI), Finland

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Software

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Summary

`aiphoria` is an open-source Python package for dynamic material flow analysis (DMFA) of wood. It can be used to follow wood from harvest through sawmills and other processing to final products such as buildings, furniture, and paper, and it calculates how much carbon these products hold over time and when it is released at end of life. Users describe the system and their scenarios in an Excel file, so no programming is needed. The package builds on ODYM (Pauliuk & Heeren, 2020) for the stock calculations and returns time series of flows, in-use stocks, and carbon stock changes, with interactive Sankey diagrams.

`aiphoria` was built for the wood sector, but its core does not depend on wood: wood-specific settings, such as carbon content and product lifetimes, are input data and can be replaced for any other material.

Statement of need

Wood is both a material and a temporary carbon store. The carbon taken up by trees stays stored in harvested wood products for as long as they remain in use, from a few years for paper to several decades for construction timber and wood panels, and is released when the products are burned or decay at end of life. Assessing the climate role of the forest-based bioeconomy therefore means following how wood flows through the economy and how long carbon stays in products before it is released. This question is central to current EU forest and climate policy (Verkerk et al., 2022).

Dynamic material flow analysis gives the basis for this (Brunner & Rechberger, 2016): it tracks wood and carbon year by year and keeps each process in balance, so harvest, products in use, and end-of-life flows are linked in one account. The main challenge is the data. Production and trade statistics are reported as absolute amounts (m^3 or tonnes), while end uses are often available only as shares, for example the percentage of sawnwood used in construction. Combining the two usually requires custom preprocessing that is hard to repeat across studies.

`aiphoria` is made for forest economists, bioeconomy analysts, and experts preparing greenhouse gas reports for land use, land-use change and forestry (LULUCF), who often work in spreadsheets rather than code. It takes absolute and relative flows together, calculates the unknown flows while keeping the mass balance, and follows the carbon in each product stock, without custom scripting.

State of the field

General DMFA frameworks such as ODYM (Pauliuk & Heeren, 2020) and flodym (Dürrwächter et al., 2026) provide well-tested methods for stocks and flows, but the user must build each system in code and first bring all data into one consistent format. `aiphoria` reuses ODYM's

dynamic stock model and adds what wood-sector studies need: a spreadsheet interface, a solver for mixed absolute and relative flows, and carbon stock accounting.

Carbon in harvested wood products (HWP) is usually estimated with the approach of the Intergovernmental Panel on Climate Change (IPCC) guidelines (IPCC, 2019). The default method starts from production statistics for three product groups (sawnwood, wood-based panels, and paper and paperboard) and assumes that each group loses carbon at a fixed rate (default half-lives of 35, 25, and 2 years). It is simple and needs little data, but it does not follow products to their end uses or capture recycling and circularity. Many models go further, and the review by Brunet-Navarro et al. (2016) shows that they differ widely in product categories, lifetimes, and end-of-life options. Studies that combine the IPCC approach with mass-balanced material flow analysis find that following end uses can give larger carbon pools than the default method (Jasinevičius et al., 2018), but these studies were not released as reusable software. Forest carbon models such as CBM-CFS3 (Kurz et al., 2009) include an HWP pool, but they start from forest inventory data, and product accounting is only a small part of a larger forest model.

Market models such as TiMBA (Morland et al., 2025) answer a different question, namely how wood flows respond to prices and policy; their results can serve as input to `aiphoria`.

What has been missing is a reusable tool that starts from production and trade statistics, follows wood to its end uses and back through recycling, and keeps mass and carbon balanced over time. Compared with the tools above, `aiphoria` adds:

- absolute amounts and shares solved together in one mass-balanced system
- in-use stocks with product lifetime distributions, calculated with ODYM
- carbon stocks and annual CO2 removals for each product stock and age cohort
- scenarios defined in Excel, with constrained and unconstrained solver modes
- interactive Sankey diagrams for every year and a network graph of the system

Software design

`aiphoria` has three layers (Figure 1). The `DataProvider` reads the Excel file; the `DataChecker` checks the data, fills in missing years, and builds the baseline and alternative scenarios; and the `FlowSolver` calculates flows and stocks year by year. Results are saved as Excel and CSV files, and the Sankey diagrams and network graph as HTML files.

The whole model, including lifetimes, carbon contents, and scenarios, is described in one Excel file. This limits what the user can express compared with code, but keeps the tool open to analysts who do not program and makes a model easy to share and review.

Each flow is given either as an absolute amount or as a share of the remaining outflow of its source process. For every year, the solver starts from the known absolute flows and calculates the rest so that the inputs and outputs of each process balance. Flows into in-use stocks are passed to ODYM's dynamic stock model, which uses a lifetime distribution to calculate the stock and its outflow by age cohort. These outflows enter the flow network again in the next year, for example as recycled wood or as wood for energy. Reusing ODYM means the stock calculations rely on a tested method.

Scenarios change selected flows over time, for example decreasing a flow share by 50% over five years. In the constrained mode, the default, a change that the data cannot supply stops the run, and the model reports the largest possible change. In the unconstrained mode the change is always applied and virtual flows fill the gap, which suits exploratory work. Virtual flows also close unreported imbalances in the input data.

`aiphoria` models flows and stocks along one dimension only: time. Location, processing stage, and carbon content are attributes of processes and flows, not separate dimensions as in ODYM and flodym. This keeps the input simple and is enough for national and regional wood-flow

studies, but it limits analyses that need to split results by, for example, region and product type at once.

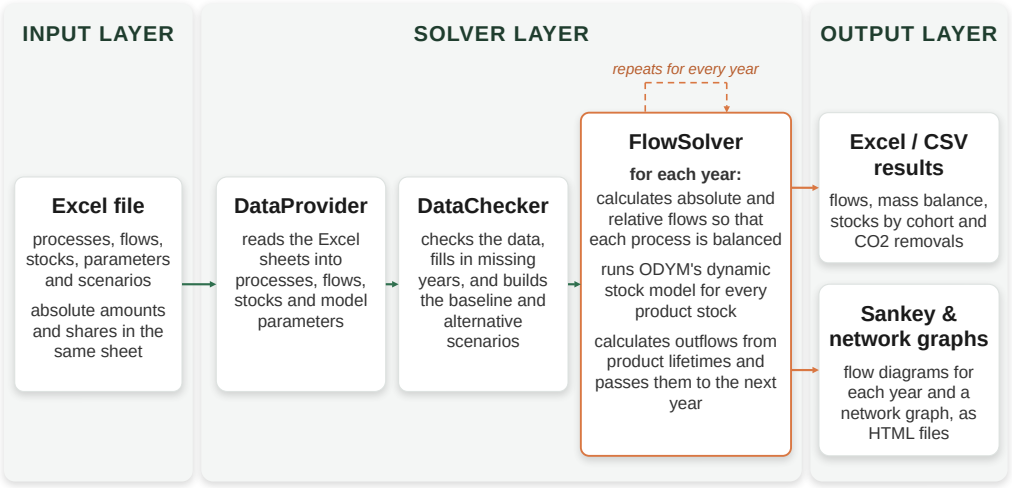


Figure 1: The three layers of an `aiphoria` run, from the Excel input to the results.

A `pytest` suite runs on every push through GitHub Actions and covers mixed flows, mass balance, both solver modes, carbon calculations, and a full run of the example. The package is on PyPI (`pip install aiphoria`).

Illustrative example

`aiphoria` comes with an example scenario that runs with two lines of code:

```
from aiphoria.example import run_example
run_example(remove_existing_output_dir=True)
```

The example uses illustrative values for a sawnwood chain in one country from 2021 to 2030 (Table 1). Sawmilling turns roundwood into sawnwood and residues. Sawnwood is imported, exported, and used in construction (mean lifetime 10 years) and furniture (5 years). At end of life, 40% of the construction wood returns to sawmilling as recycled wood, and the rest, together with all furniture, goes to incineration. An alternative scenario reduces sawmilling residues by 50% between 2025 and 2030.

Table 1: Flows of the example scenario as entered in the Excel input (year 2021). Values are illustrative.

Source	Target	Value	Unit
Industrial roundwood	Sawmilling	60	Mm^3
Sawmilling	Residues	10	Mm^3
Sawmilling	Sawnwood	100	%
Sawnwood (import)	Sawnwood	10	Mm^3
Sawnwood	Sawnwood (export)	20	Mm^3
Sawnwood	Construction	60	%
Sawnwood	Furniture	40	%
Construction	Sawmilling	40	%
Construction	Incineration	60	%
Furniture	Incineration	100	%

Figure 2 shows the resulting flows. In 2021 (a), almost all wood entering construction and furniture stays in use. By 2030 (b), the first furniture and construction cohorts have reached end of life, so flows to incineration and a recycling loop from construction back to sawmilling appear.

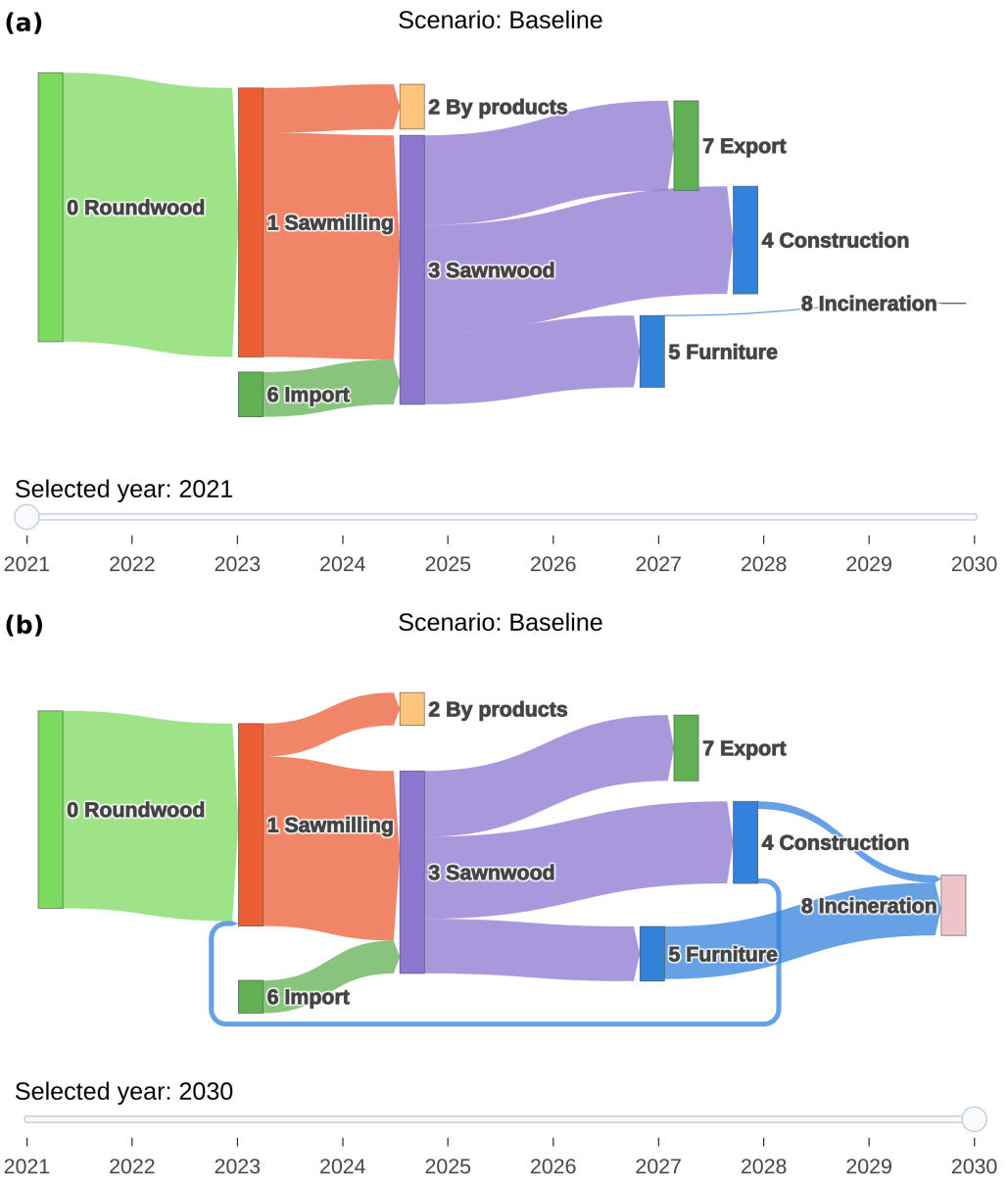


Figure 2: Sankey diagram of the example scenario in (a) 2021 and (b) 2030, selected with the year slider.

The carbon stocks are also saved to Excel (Table 2). Furniture has a short lifetime (5 years), so its stock stops growing at about 20 Mt C once as much carbon leaves as enters. Construction has a longer lifetime (10 years), so its stock keeps growing until 2030.

Table 2: Carbon stock in the in-use product stocks of the example scenario (baseline), in million tonnes of carbon (Mt C), from the `Total_stock` sheet of the output workbook.

Stock	2021	2024	2027	2030
Construction	5.4	21.6	37.8	53.2
Furniture	3.6	14.3	19.7	20.0

Research impact statement

`aiphoria` has supported two peer-reviewed studies. It was used to map wood material flows across the EU forest sector (Orfanidou et al., 2026) and to reconstruct the Italian forest-wood value chain (Khan et al., 2026). The package was developed in the Horizon Europe ForestPaths project, is in active use at the European Forest Institute (EFI), and has had more than sixteen releases under an MIT licence, with documentation in the project wiki.

AI usage disclosure

The authors used Anthropic's Claude models (Claude Sonnet 5 and Claude Opus 5.5) while preparing this paper. All scientific content, design decisions, and the software itself were produced by the authors, who reviewed, edited, and verified every AI-assisted change.

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References

Brunet-Navarro, P., Jochheim, H., & Muys, B. (2016). Modelling carbon stocks and fluxes in the wood product sector: A comparative review. *Global Change Biology*, *22*(7), 2555–2569. https://doi.org/10.1111/gcb.13235

Brunner, P. H., & Rechberger, H. (2016). *Handbook of material flow analysis: For environmental, resource, and waste engineers* (2nd ed.). CRC Press. https://doi.org/10.1201/9781315313450

Dürrwächter, J., Hosak, M., Weiss, B., & Ueckerdt, F. (2026). Flodym: A Python package for dynamic material flow analysis. *Journal of Open Source Software*, *11*(119), 10105. https://doi.org/10.21105/joss.10105

IPCC. (2019). Harvested wood products. In *2019 refinement to the 2006 IPCC guidelines for national greenhouse gas inventories, Volume 4: Agriculture, forestry and other land use*. Intergovernmental Panel on Climate Change. https://www.ipcc-nggip.iges.or.jp/public/2019rf/vol4.html

Jasinevičius, G., Lindner, M., Cienciala, E., & Tykkyläinen, M. (2018). Carbon accounting in harvested wood products: Assessment using material flow analysis resulting in larger pools

compared to the IPCC default method. *Journal of Industrial Ecology*, *22*(1), 121–131. https://doi.org/10.1111/jiec.12538

Khan, M. T., Masiero, M., Pettenella, D., Andrighetto, N., Weimar, H., & Glasenapp, S. (2026). Mapping and analysing wood flows across the Italian forest-based wood value chain: A material flows analysis. *European Journal of Wood and Wood Products*, *84*(3), 85. https://doi.org/10.1007/s00107-026-02428-y

Kurz, W. A., Dymond, C. C., White, T. M., Stinson, G., Shaw, C. H., Rampley, G. J., Smyth, C., Simpson, B. N., Neilson, E. T., Trofymow, J. A., Metsaranta, J., & Apps, M. J. (2009). CBM-CFS3: A model of carbon-dynamics in forestry and land-use change implementing IPCC standards. *Ecological Modelling*, *220*(4), 480–504. https://doi.org/10.1016/j.ecolmodel.2008.10.018

Morland, C., Schier, F., Tandetzki, J., & Honkomp, T. (2025). TiMBA (timber market model for policy-based analysis). *Journal of Open Source Software*, *10*(115), 8034. https://doi.org/10.21105/joss.08034

Orfanidou, T., Martinez, G. E., Järvikylä, J., Verkerk, P. J., & Cardellini, G. (2026). Mapping wood flows to final product and energy uses in 2021 in the European Union. *Resources, Conservation & Recycling Advances*, *30*, 200331. https://doi.org/10.1016/j.rcradv.2026.200331

Pauliuk, S., & Heeren, N. (2020). ODYM—An open software framework for studying dynamic material systems: Principles, implementation, and data structures. *Journal of Industrial Ecology*, *24*(3), 446–458. https://doi.org/10.1111/jiec.12952

Verkerk, P. J., Delacote, P., Hurmekoski, E., Kunttu, J., Matthews, R., Mäkipää, R., Mosley, F., Perugini, L., Reyer, C. P. O., Roe, S., & Trømborg, E. (2022). *Forest-based climate change mitigation and adaptation in Europe* (No. 14; From Science to Policy). European Forest Institute. https://doi.org/10.36333/fs14